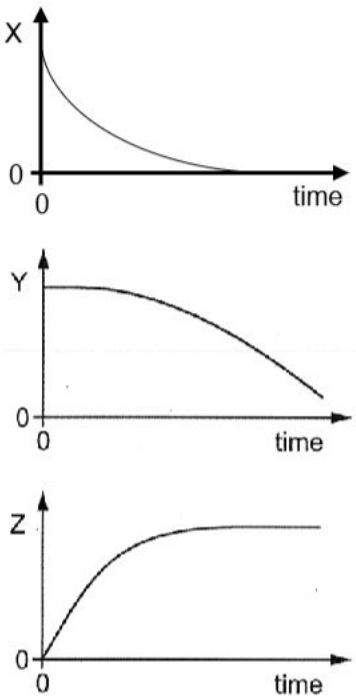
2 An object is dropped at time $t = 0$ s from a tall building. Air resistance is significant.
Three graphs are plotted against time:

- the height of the object above the ground
- the speed of the object
- the magnitude of the resultant force on the object.



What are the quantities X, Y and Z?

	height of the object above the ground	speed of the object	magnitude of the resultant force on the object
A	X	Y	Z
B	X	Z	Y
C	Z	Y	X
D	Y	Z	X

Ans: D

Graph Y represents the height-time graph. The object must start from a certain height above the ground. The gradient of the graph gives the speed, which increases from zero to a constant terminal speed.

Graph Z represents the speed-time graph. The speed must start from zero and reach a constant terminal speed.

Graph X represents the resultant force-time graph. At first, the speed of the object is zero, so there so no air resistance. The resultant force is therefore equal to the weight. But as the object falls and speeds up, the air resistance rises, so the resultant force drops until terminal speed is reached, when air resistance = weight.